BALA JEE SONI

Mechanical Engineering • Design & Simulation • Robotics
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SUMMARY

Final-year B.Tech Mechanical Engineering student (graduating June 2027) with strong SolidWorks CAD (CSWA certified) and motion simulation skills, working knowledge of ANSYS finite element analysis (FEA), materials testing, and intermediate Python for engineering analysis. Founded BIET's first robotics and sustainability club (40+ members). Seeking a full-time entry-level engineering role in mechanical design, simulation or robotics.

EDUCATION

Bundelkhand Institute of Engineering & Technology (BIET), Jhansi | *B.Tech, Mechanical Engineering* 2023 – June 2027

- CGPA: 7.01 | Coursework: Fluid Mechanics, Thermodynamics, Machine Design, Materials Engineering, Manufacturing

PROJECTS

Final Year Project: Six-Wheel Rocker-Bogie Rover | *Dept. of Mechanical Engineering, BIET* In progress

- Developed the design plan and roadmap for a 6-wheel rocker-bogie rover, backed by research papers.
- Designed the analysis plan covering IMU-based stability analysis, power management and motor configuration testing.

Analysis of Planetary Gear Systems | *SolidWorks, Python* Sep 2025

- Designed and assembled a planetary gear system in SolidWorks CAD; verified kinematics with a Motion Study simulation.
- Developed a Python tool to calculate gear geometry, rotational speeds, torque and forces, and to validate design constraints.

Comparative Analysis of Mechanical Properties of Mild Steel After Heat Treatment Aug – Sep 2024

- Executed annealing, normalising and water quenching on mild steel specimens, creating 4 material conditions.
- Analysed the conditions through 3 tests (hardness, impact and tensile testing) to evaluate heat treatment effects.

EXPERIENCE

Railway Workshop, Gorakhpur | *Industrial Training, Mechanical Workshop* June 2025 (4 weeks)

- Examined material handling equipment and its workshop applications during 4 weeks of mechanical workshop training.

SKILLS

Strong: SolidWorks, CSWA certified (CAD part modelling, assemblies, Motion Study simulation)
Intermediate: Python (engineering calculations, NumPy, pandas)
Familiar: CAD, Simulation & Testing: ANSYS (Finite Element Analysis), Onshape, FreeCAD, MATLAB/Simulink, Ultimaker Cura (3D printing), materials testing (tensile, hardness, impact)
Familiar: Programming & Electronics: C, C++, Arduino Uno, ESP32, KiCad, ROS2, Gazebo, Webots, OpenCV, YOLO, MediaPipe, Hugging Face
Professional: Problem Solving, Project Management, Cross-functional Collaboration, Technical Documentation
Currently Learning: Gazebo, ROS2, KiCad PCB design
Languages: Hindi (native), English (working proficiency)

LEADERSHIP & CLUBS

Solarpunk Corps (SPC) | *Founder & General Secretary* Jan 2026 – Present

- Founded BIET's first robotics and sustainability club (@we.the.spc) under the Mechanical Engineering Department, open to all departments; built a team of 40+ active members.
- Led and project-managed 2 workshops (Horizon 1.0: 3 days, Horizon 2.0: 4 days), reaching 300+ participants; 100 shortlisted candidates built projects in electronics, AI/ML, robotics and sustainability.
- Coordinated cross-functional student teams (25–30 organisers) and prepared curriculum, documentation and outreach material.

Alumni Association, BIET Jhansi | *Advisory Board Member* Aug 2025 – Present

- Developed content in the Content Development Team, then moved to the Advisory Board linking students and alumni.

BIS Student Chapter | *Member* Aug 2025 – Present

- Built knowledge of robotics and engineering standards through chapter sessions.

Mercenary Motorsports Club (MMC), BIET Jhansi | *Member* Aug 2024 – Jul 2025

- Developed working skills in SolidWorks CAD and ANSYS finite element analysis (FEA) basics through club workshops.

The Drone Learners Club, BIET Jhansi | *Member* Dec 2024 – Jan 2025

- Built a working Styrofoam RC aircraft (Oct 2025) and analysed RC flight motions, BLDC motors, ESCs and gyroscopes.

CERTIFICATIONS & WORKSHOPS

Certified SOLIDWORKS Associate (CSWA), Dassault Systèmes SOLIDWORKS October 2026

AI & Robotics Club Online Program, Technova Society, Sharda University 31 Jan – 1 Feb 2026

- Completed the two-day virtual workshop "Building the Smart World with IoT".

Python for Beginners, Simplilearn January 2025